

```

nop = 4
nelem = 4
ngl = 5
nq = 6
npoin_cg = 17
npoin_dg = 20
u = 2

```

Legendre Gauss Lobatto Weights (wgl):
[0.1, 0.544444, 0.711111, 0.544444, 0.1]

Quadrature Weights (wnq):
[0.0666667, 0.378475, 0.554858, 0.554858, 0.378475, 0.0666667]

```
CG Coordinates (coord_cg):
[-1, -0.913663, -0.75, -0.586337, -0.5, -0.5, -0.413663, -0.25, -0.0863366, 0, 0, 0.0863366, 0.25, 0.413663]
```

```
CG Connectivity (intma_cg):  
[0]
```

```
CG Periodicity (periodicity_cg):  
[0]
```

DG Periodicity (periodicity_dg):
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19]

```
Lagrange Basis Derivative (dpsi):
[-5, -2.15493, 0.294142, 0.0287337, -0.167949, 0.5;
 6.7565, 1.54337, -1.98555, -0.0351799, 0.477352, -1.41016;
 -2.66667, 0.920956, 1.68496, -1.68496, -0.920956, 2.66667;
 1.41016, -0.477352, 0.0351799, 1.98555, -1.54337, -6.7565;
 -0.5, 0.167949, -0.0287337, -0.294142, 2.15493, 5]
```

```
Mass Matrix (Me):  
[0.00222222, 0.00648148, -0.00740741, 0.00648148, -0.00277778;  
 0.00648148, 0.120988, 0.017284, -0.0151235, 0.00648148;  
 -0.00740741, 0.017284, 0.158025, 0.017284, -0.00740741;  
 0.00648148, -0.0151235, 0.017284, 0.120988, 0.00648148;  
 -0.00277778, 0.00648148, -0.00740741, 0.00648148, 0.0222222;  
 0.0222222, 0.00648148, -0.00740741, 0.00648148, -0.00277778;  
 0.00648148, 0.120988, 0.017284, -0.0151235, 0.00648148;  
 -0.00740741, 0.017284, 0.158025, 0.017284, -0.00740741;  
 0.00648148, -0.0151235, 0.017284, 0.120988, 0.00648148;  
 -0.00277778, 0.00648148, -0.00740741, 0.00648148, 0.0222222;  
 0.0222222, 0.00648148, -0.00740741, 0.00648148, -0.00277778;  
 0.00648148, 0.120988, 0.017284, -0.0151235, 0.00648148;  
 -0.00740741, 0.017284, 0.158025, 0.017284, -0.00740741;  
 0.00648148, -0.0151235, 0.017284, 0.120988, 0.00648148;
```

```
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, -2;
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0;
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0;
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0;
0, 0, 0, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0;
0, 0, 0, 0, -2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0;
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0;
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0;
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0;
0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0;
0, 0, 0, 0, 0, 0, 0, 0, -2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0;
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0;
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0;
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0, 0, 0, 0;
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, -2, 0, 0, 0, 0, 0;
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0;
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0;
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0;
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 21]
```

[illegible][illegible]

```
[ -1, -1.3513, 0.533333, -0.282033, 0.1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2;  
1.3513, -3.95517e-16, -1.90092, 0.831653, -0.282033, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0;  
-0.533333, 1.90092, 3.05311e-16, -1.90092, 0.533333, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0;  
0.282033, -0.831653, 1.90092, 0, -1.3513, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0;  
-0.1, 0.282033, -0.533333, 1.3513, -1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0;  
0, 0, 0, 0, 2, -1, -1.3513, 0.533333, -0.282033, 0.1, 0, 0, 0, 0, 0, 0, 0, 0, 0;  
0, 0, 0, 0, 0, 1.3513, -3.95517e-16, -1.90092, 0.831653, -0.282033, 0, 0, 0, 0, 0, 0, 0, 0, 0;  
0, 0, 0, 0, 0, -0.533333, 1.90092, 3.05311e-16, -1.90092, 0.533333, 0, 0, 0, 0, 0, 0, 0, 0, 0;  
0, 0, 0, 0, 0, 0.282033, -0.831653, 1.90092, 0, -1.3513, 0, 0, 0, 0, 0, 0, 0, 0, 0;  
0, 0, 0, 0, 0, -0.1, 0.282033, -0.533333, 1.3513, -1, 0, 0, 0, 0, 0, 0, 0, 0, 0;  
0, 0, 0, 0, 0, 0, 0, 0, 2, -1, -1.3513, 0.533333, -0.282033, 0.1, 0, 0, 0, 0, 0;  
0, 0, 0, 0, 0, 0, 0, 0, 0, 1.3513, -3.95517e-16, -1.90092, 0.831653, -0.282033, 0, 0, 0, 0, 0
```